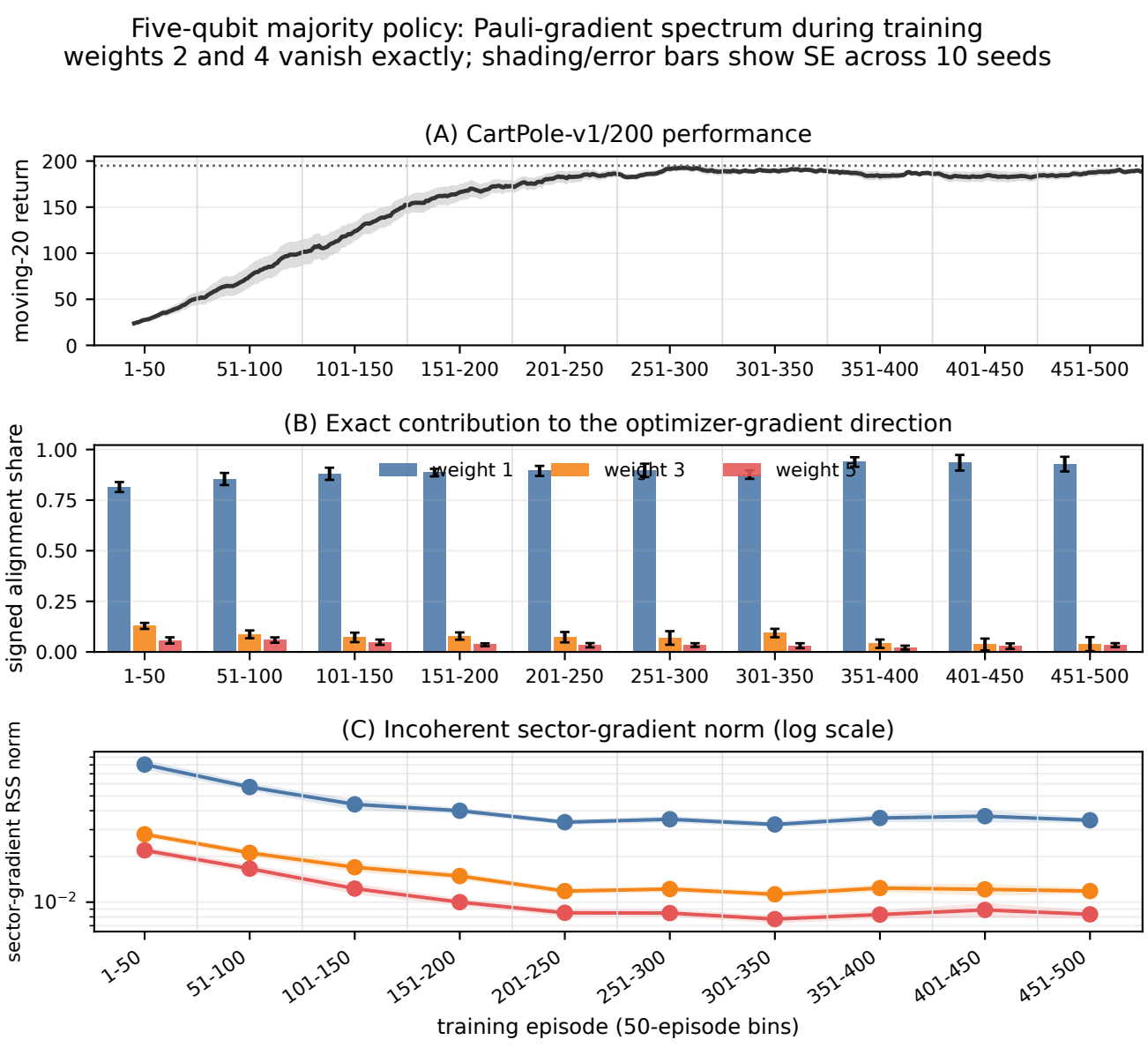
Five-qubit majority policy: Pauli-gradient spectrum during training
weights 2 and 4 vanish exactly; shading/error bars show SE across 10 seeds
(A) CartPole-v1/200 performance
moving-20 return
0
50
100
150
200
1-50
51-100
101-150
151-200
201-250
251-300
301-350
351-400
401-450
451-500
(B) Exact contribution to the optimizer-gradient direction
signed alignment share
0.00
0.25
0.50
0.75
1.00
weight 1
weight 3
weight 5
(C) Incoherent sector-gradient norm (log scale)
sector-gradient RSS norm
10^{-2}
training episode (50-episode bins)